

MAC-RRG: Iterative Multi-Agent Collaboration for X-ray Radiology Report Generation

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Abstract—Despite the remarkable progress of LLM-based and knowledge graph-augmented Radiology Report Generation (RRG) methods, existing techniques still suffer from inherent defects. Conventional LLM-only models lack structured medical prior knowledge, resulting in frequent medical hallucinations and low diagnostic interpretability. Current knowledge graph-enhanced schemes adopt static one-round knowledge fusion with single-source knowledge, incapable of dynamic knowledge updating according to generation feedback. This paper proposes a novel Multi-Agent Collaborative iterative framework for X-ray Radiology Report Generation, termed MAC-RRG. Inspired by multi-agent technology, our framework constructs a closed-loop optimization paradigm based on task decoupling and collaborative reasoning. Specifically, the framework first generates a preliminary radiology report from input X-ray images via a vision encoder and a basic LLM. Subsequently, a multimodal knowledge graph (MM-KG) agent mines structured disease correlation and anatomical knowledge from medical knowledge graphs, while an auxiliary knowledge agent extracts unstructured domain knowledge from public medical databases. The multi-source knowledge acquired by dual agents is fused and embedded to guide the LLM in iteratively refining the initial report. Extensive quantitative and qualitative experiments on mainstream X-ray RRG datasets, including IU X-ray, MIMIC, and CheXpert Plus, fully verify the superiority of our proposed method. The source code and pre-trained models have been released on https://github.com/Event-AHU/Medical_Image_Analysis

Index Terms—Radiology Report Generation, Multi-Agent Collaboration, Large Language Model, Knowledge-guided Learning, Iterative Learning

I. INTRODUCTION

CHEST X-ray is one of the most widely adopted diagnostic imaging modalities in routine clinical screening, emergency diagnosis and respiratory disease surveillance. Radiology Report Generation (RRG) [1], which aims to automatically generate professional, accurate diagnostic descriptions corresponding to X-ray images, has attracted extensive research attention in the community of medical vision and large multimodal models. Manually writing radiology reports imposes a heavy workload on radiologists, often leading to prolonged waiting times for patients and increased risks of oversight or inconsistent descriptions caused by clinician

fatigue. Automating X-ray RRG can effectively alleviate the labor burden of radiologists, streamline clinical workflows, and serve as an auxiliary diagnostic tool to improve the accessibility of radiological services, especially in regions facing shortages of professional radiology practitioners. Nevertheless, reliable X-ray RRG remains a challenging task: the model is required to precisely identify subtle radiographic lesions, capture interrelations among multiple imaging findings, and generate clinically consistent, logically rigorous diagnostic texts that conform to standardized radiological conventions [2].

In the deep learning era, the performance of X-ray RRG has been continuously advanced. Early research mainly built upon convolutional neural networks (CNNs) [3], such as the multi-task co-attention model proposed by Jing et al. [4], TieNet [5], which jointly performs thoracic disease classification and radiology report generation, and HRGR-Agent [6], which combines template retrieval with free-text generation. These methods typically adopt an encoder-decoder paradigm, where CNNs extract visual features from X-ray images, followed by recurrent networks like LSTM [7] to generate descriptive texts. Later, Transformer architectures [8] gradually became the mainstream backbone for RRG tasks, represented by R2Gen [9], which introduces a relational memory mechanism to enhance the modeling of report generation patterns and long-range dependencies in lengthy texts; R2GenCMN [10] further employs a cross-modal memory network to explicitly model the correspondence between radiographic image regions and textual descriptions. Building upon these methods, PPKED [11] explores and distills prior and posterior knowledge to alleviate visual and linguistic biases caused by the imbalanced distribution of normal and abnormal cases; ME-Transformer [12] introduces multiple learnable expert tokens, enabling different experts to attend to complementary image regions and generate reports by aggregating their predictions.

Benefiting from the self-attention mechanism, Transformer-based models are capable of modeling long-range dependencies among imaging regions and output tokens, yielding clearer correspondence between visual lesions and textual descriptions. With the rapid advancement of pre-trained large multimodal models [13] in recent years, the quality of automatically generated X-ray radiology reports has achieved substantial improvements, as exemplified by R2GenGPT [14], which employs a lightweight visual alignment module to map radiographic image features into the word embedding space of a frozen large language model, thereby improving report generation capabilities while reducing the number of trainable parameters; R2GenCSR [15], which retrieves positive and

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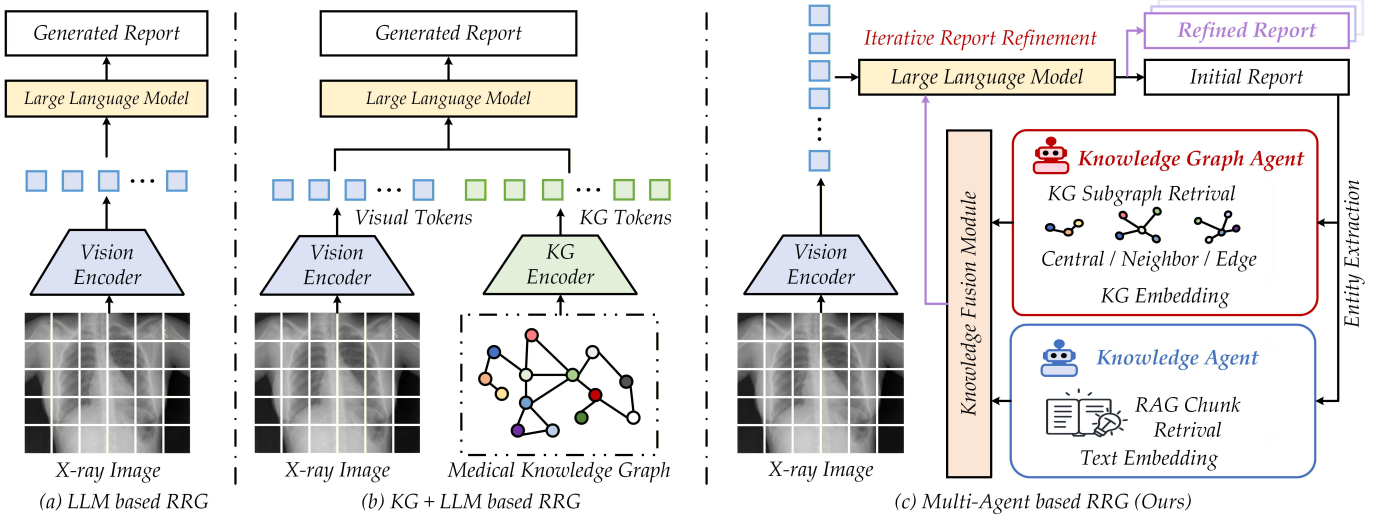


Fig. 1. Comparison between existing RRG frameworks and our newly proposed multi-agent RRG based model. (a) The baseline LLM-based medical report generation paradigm only fuses visual features extracted from raw X-ray images without incorporating structured prior medical knowledge. (b) Knowledge graph-enhanced LLM-based medical report generation models feed KG tokens encoded from medical knowledge graphs together with visual tokens into large language models. (c) We propose a novel iterative multi-agent collaboration framework for X-ray radiology report generation, termed MAC-RRG, which leverages collaborative multi-agent knowledge mining and fusion on initial radiology reports to iteratively enhance report quality.

negative contextual samples related to the current image to provide supplementary semantic references and jointly feeds visual features, retrieved contexts, and textual prompts into a large language model, thereby enhancing the discriminative capability of visual representations and improving report generation quality; and MambaXray-VL [16], which adopts a Mamba-based visual backbone and constructs a multi-stage pre-training framework integrating self-supervised autoregressive learning, image-report contrastive learning, and supervised fine-tuning to enhance the visual representations of X-ray images and cross-modal alignment capabilities. By leveraging large-scale image-text pre-training data, these models establish stronger cross-modal alignment between radiographic visual content and professional radiological language, further narrowing the performance gap between machine-generated narratives and manually written clinical reports.

Despite the aforementioned progress, we believe the following issues still limit these works: **(1).** The baseline LLM-based medical report generation paradigm only fuses visual features extracted from raw X-ray images without incorporating structured prior medical knowledge, as shown in Fig. 1 (a). As a result, the model cannot effectively leverage domain-specific medical knowledge such as disease correlations and anatomical priors. When interpreting complex imaging findings, it is prone to medical hallucinations, and its diagnostic reasoning lacks interpretability. **(2).** Knowledge graph-enhanced LLM-based medical report generation models feed KG tokens encoded from medical knowledge graphs together with visual tokens into large language models, as shown in Fig. 1 (b), which mitigates the problem of insufficient external knowledge to a certain extent. However, the embedded knowledge graph features are integrated statically in a one-off manner during forward inference, making it impossible to dynamically adjust and iteratively retrieve required knowledge conditioned

on preliminarily generated reports. Moreover, such methods only rely on structured graph knowledge from knowledge graphs, without supplementary unstructured textual knowledge including medical literature and clinical guidelines. **(3).** Both aforementioned approaches adopt a one-shot report generation pipeline without a feedback loop for iterative refinement. Errors, including entity omissions and biased descriptions of imaging findings in initial reports, cannot be continuously verified and corrected using external medical knowledge.

To tackle the inherent drawbacks of *static knowledge fusion*, *single-round generation*, and *insufficient multi-source knowledge utilization* in existing RRG methods, we are inspired by the emerging multi-agent technology [17], [18]. Different from conventional static embedding and one-shot inference pipelines, the multi-agent collaboration mechanism possesses distinct advantages of task decoupling, specialized functional division, dynamic interactive reasoning, and iterative closed-loop optimization. Specifically, professionalized knowledge agents can be deployed to undertake targeted knowledge mining tasks separately, breaking the limitation of single structured knowledge sources. Meanwhile, the interactive iteration mechanism between agents and the generative LLM enables dynamic knowledge retrieval and real-time correction guided by initial report feedback, rather than passive one-time knowledge fusion. This hierarchical collaborative and iterative optimization paradigm perfectly compensates for the lack of interpretability, static knowledge fusion, and non-iterative generation in previous RRG frameworks, providing a feasible solution to the above-mentioned bottlenecks.

In this paper, we propose a novel iterative multi-agent collaboration framework for X-ray radiology report generation, termed MAC-RRG, which leverages collaborative multi-agent knowledge mining and fusion on initial radiology reports to iteratively enhance report quality. Given the X-ray image, we first partition it into multiple image patches and project them

into visual tokens. A vision encoder is adopted to further enhance the features, and a Large Language Model (LLM) is used to produce an initial radiology report. Then, we extract the entities from the report and mine the key centrals/neighbors/edges from the multi-modal X-ray knowledge graph using the MM-KG agent, and extract relevant domain knowledge from existing public medical knowledge bases [19] using another knowledge agent. Such knowledge is encoded into token embeddings and aggregated by the knowledge fusion module. The fused features are subsequently fed into the large language model to achieve iterative radiology report refinement. The comparison between existing radiology report generation models and our newly proposed framework can be found in Fig. 1. An overview of our MAC-RRG can be found in Fig. 2.

To sum up, the main contributions of this paper can be summarized as follows:

- We propose a novel iterative multi-agent collaboration framework for X-ray radiology report generation, termed MAC-RRG, which enables multiple agents to collaboratively perform supplementary knowledge mining and fusion over initial radiology reports for iterative report enhancement.
- We propose a multimodal knowledge graph agent, a knowledge agent, and a knowledge fusion module for X-ray radiology report generation, which jointly enable high-quality iterative refinement of radiology reports.
- Extensive experiments on multiple mainstream X-ray radiology report datasets demonstrate the efficacy of our method. In comparison with conventional large-model approaches and knowledge graph-enhanced methods, the proposed model obtains improved interpretability and enhanced capabilities for disease diagnosis and analysis.

The rest of this paper is organized as follows: We give a brief review of the most related works in Section II, and describe our framework in Section III, with a focus on the overview of our framework, initial report generation and entity extraction, multi-agent based knowledge fusion, and the loss function. In Section IV, we conduct experiments to validate the effectiveness in both quality and quantity analysis. We conclude this paper in Section V.

II. RELATED WORKS

A. Radiology Report Generation

RRG aims to generate structurally complete and clinically factual free-text reports from medical images such as chest X-rays. Early studies typically followed the paradigm of image captioning by combining CNN-based visual encoders with recurrent decoders or hierarchical language models [4], [5], where co-attention mechanisms were employed to localize abnormal regions and generate multi-sentence reports. Subsequently, Transformer architectures and memory mechanisms have been widely adopted to model inter-sentence dependencies and the distribution of medical terminology in long-form reports. For example, R2Gen [9] introduces a memory-driven Transformer to record key semantic information during the generation process, while R2GenCMN [10] further employs cross-modal shared memory to explicitly model the alignment

between visual and textual representations. These methods have substantially advanced end-to-end RRG; however, their language generation modules typically rely on training from scratch or large-scale fine-tuning on task-specific data. In long-text and fact-sensitive scenarios such as medical report generation, they remain prone to abnormality omission, vision-language mismatch, and template-like report generation.

In recent years, large language models (LLMs) have been introduced into RRG to leverage their strong language modeling ability, medical terminology organization, and instruction-following capability. R2GenGPT [14] aligns visual features with the word embedding space of a frozen LLM, enabling efficient report generation by training only a lightweight alignment module. RaDialog [20] further integrates visual features, structured pathological findings, and instruction data to support both report generation and interactive radiology-oriented dialogue. These LLM-based RRG methods demonstrate that LLMs can serve as powerful generators. Nevertheless, relying solely on visual alignment is still insufficient to ensure that each key entity is supported by adequate external medical evidence.

B. Medical Knowledge Graphs

Medical knowledge-enhanced methods focus on explicitly incorporating anatomical structures, disease concepts, and clinical relationships into radiology report generation models. KERP [21] decomposes medical image report generation into three stages, namely abnormality graph learning, retrieval, and paraphrasing, and provides early evidence for the value of explicit medical knowledge in improving report accuracy. PP-KED [11] further explores posterior and prior knowledge from abnormal regions, medical knowledge graphs, and historical reports, and integrates such knowledge into report generation through knowledge distillation. KiUT [22] employs a symptom graph and a knowledge distiller to facilitate multi-level visual-textual interaction in a U-Transformer, while KARGEN [23] introduces a disease-related knowledge graph within a frozen LLM framework to activate more relevant knowledge about thoracic diseases.

Structured radiology resources also provide an important foundation for knowledge-enhanced RRG. RadGraph [24] annotates and extracts clinical entities and their relations from free-text radiology reports, while Chest ImaGenome [25] represents chest X-rays as anatomy-centered scene graphs, indicating that clinical facts in radiology reports are naturally suited to entity-relation structured representations. Collectively, these studies suggest that report quality depends not only on whether a specific disease is correctly identified, but also on whether the relationships between diseases and anatomical locations, attributes, temporal changes, and adjacent findings are clinically consistent.

C. Retrieval-Augmented Generation and Agentic Workflows

Retrieval-Augmented Generation (RAG) [26] combines parametric generative models with non-parametric external memory, enabling models to generate more specific text by leveraging updatable and traceable external evidence. In chest

X-ray report generation, Retrieval-Augmented Chest X-Ray Report Generation [27] retrieves candidate report texts using multimodally aligned embeddings and feeds them into general-purpose GPT models for report generation. LaB-RAG [28] enhances textual retrieval with image-derived labels, while RA-RRG [29] employs clinical key phrase retrieval to reduce hallucinations and computational cost. These methods demonstrate that retrieval can compensate for medical knowledge missing or unstable in model parameters. However, most existing approaches rely on image-level or report-level similarity, and the retrieved results may not fully match the specific entities in the current sample, thereby introducing noisy evidence.

Agentic language model research further demonstrates that generation can be improved by decomposing complex tasks into intermediate reasoning, tool use, and knowledge acquisition steps. For example, ReAct [30] interleaves reasoning traces with task-specific actions, enabling language models to interact with external knowledge sources during inference. Inspired by this idea, we adopt a lightweight agentic workflow tailored to radiology report generation. Our framework instantiates role-specialized knowledge agents: an MM-KG Agent retrieves and aggregates structured entity relations from a medical knowledge graph, while a Knowledge Agent retrieves complementary textual evidence from an external medical knowledge corpus. The two knowledge streams are integrated through a knowledge fusion module to refine the initial report, enabling entity-aware, source-complementary, and clinically grounded report generation.

III. OUR PROPOSED APPROACH

A. Overview

As shown in Fig. 2, we propose an iterative multi-agent collaboration-enhanced framework for X-ray radiology report generation, termed MAC-RRG. The framework first employs a visual encoder and a large language model to generate an initial report, from which anatomical structures and disease or abnormality concepts are subsequently extracted. These clinically meaningful entities are then processed through two complementary branches. In the graph branch, local neighborhoods are constructed based on a medical knowledge graph and encoded using Bio_ClinicalBERT. In the retrieval branch, a BGE-based retriever and reranker are used to retrieve relevant textual evidence for each entity, which is likewise encoded using Bio_ClinicalBERT. Finally, the graph knowledge tokens and retrieval knowledge tokens are integrated through the Knowledge Fusion Module, concatenated with the image tokens, and projected into the hidden space of the large language model. The decoder then generates radiology reports that are more consistent with established medical standards.

B. Initial Report Generation and Entity Extraction

The visual encoder maps chest X-ray images into a sequence of image tokens that preserve radiological evidence. Each input view $\mathbf{I}$ is encoded by the Swin Transformer $f_v(\cdot)$ as follows:

$$\mathbf{V} = f_v(\mathbf{I}) \in \mathbb{R}^{N_v \times d_v}, \quad (1)$$

where $\mathbf{V}$ denotes the visual token sequence, N_v is the number of visual tokens, and d_v is the feature dimension. The visual sequence $\mathbf{V}$ is then projected into the hidden dimension of the LLM and used to autoregressively generate an initial report.

The initial report provides explicit textual anchors for selecting clinically relevant knowledge. As illustrated in the upper panel of Fig. 3, the draft report R_0 serves as the starting point for subsequent knowledge retrieval. Given the draft report R_0 , the entity extraction module first segments it into sentences, normalizes punctuation and letter case, and performs lexical matching based on a radiology entity dictionary. We retain only two types of entities, namely *anatomy* and *disorder*, since they directly describe the most essential anatomical regions and abnormal findings in chest X-ray reports. Duplicate entity mentions across sentences are then merged to form an examination-level entity set:

$$\mathcal{E} = \{(e_i, t_i)\}_{i=1}^M, \quad (2)$$

where e_i denotes the normalized entity mention, and $t_i \in \{\text{anatomy}, \text{disorder}\}$ denotes the entity type.

Using the initial report for entity selection has two advantages. First, it restricts external knowledge queries to image-conditioned clinical concepts, rather than retrieving general radiological priors. Second, it provides a shared entity set for both the graph branch and the retrieval branch, enabling structured and unstructured knowledge to remain comparable at the token level.

C. MM-KG Agent

The MM-KG Agent injects structured radiological relations into the report generation process by expanding each preliminary report entity into a local knowledge graph domain. The medical knowledge graph is stored as a collection of weighted triples. For each entity e_i , the agent searches for triples whose source or target entity matches e_i . The matched edges are ranked according to frequency counts, and the top- K_g neighboring entities are retained:

$$\mathcal{G}_i = \{(e_i, r_{ij}, n_{ij})\}_{j=1}^{K_i}, \quad (3)$$

where n_{ij} denotes the neighboring entity and r_{ij} represents the corresponding relation type. In the current setting, each entity is linked to at most ten neighboring entities, i.e., $K_g = 10$.

The central entity, neighboring entities, and relation labels are encoded using a frozen Bio_ClinicalBERT encoder. Formally, we define

$$\hat{\mathcal{G}}_i = \{(\phi(e_i), \phi(n_{ij}), \phi(r_{ij}))\}_{j=1}^{K_i}, \quad (4)$$

where $\phi(\cdot)$ denotes the Bio_ClinicalBERT text encoding function. These embeddings are subsequently fed into the Knowledge Fusion Module, where they are fused with the retrieval-based knowledge representations produced by the Knowledge Agent.

This graph encoding design aims to preserve the relational structure among anatomical regions, imaging findings, and diseases. Rather than processing each concept independently, the model is provided with local graph context, enabling it to capture clinically meaningful associations during report generation.

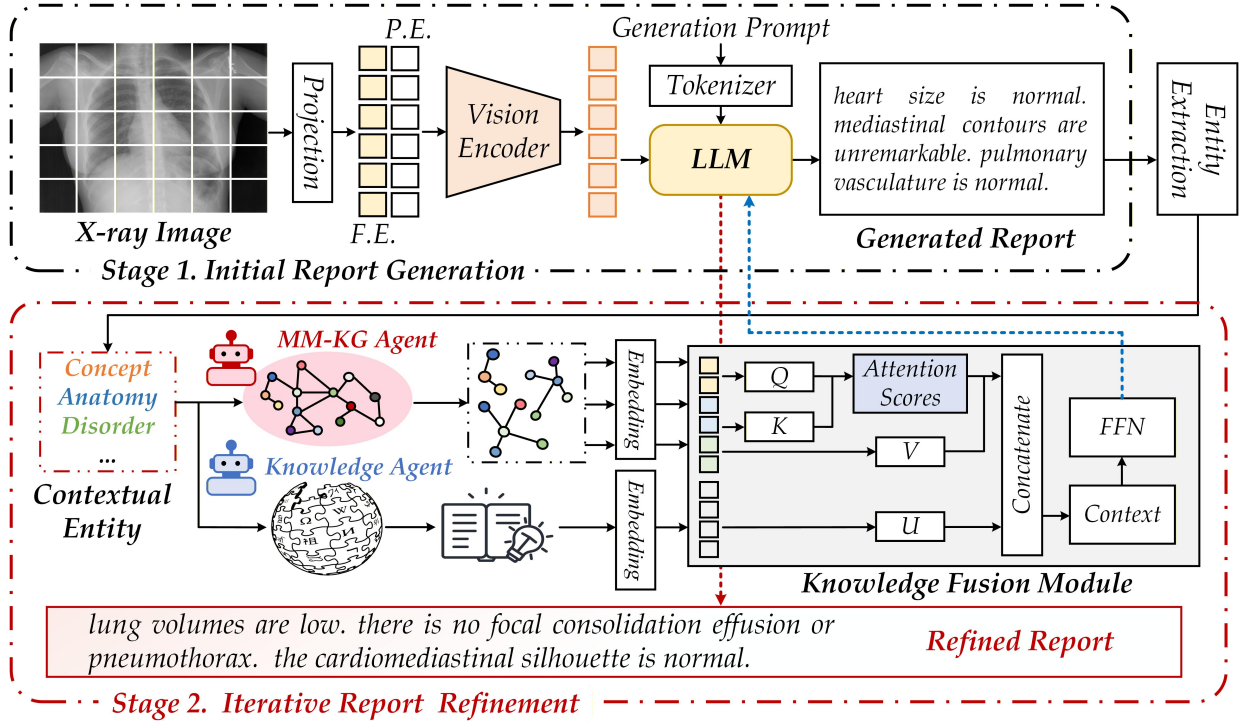


Fig. 2. An overview of our iterative multi-agent collaboration for X-ray report generation framework, termed MAC-RRG. The framework comprises two stages: (1) Initial Report Generation, in which the X-ray image is encoded into visual tokens and fed into a large language model to generate an initial radiology report. (2) Iterative Report Refinement, where the extracted entities are used to construct local neighborhoods from a medical knowledge graph and retrieve relevant textual evidence. The graph knowledge and retrieval knowledge are encoded and fused with image tokens through the Knowledge Fusion Module, and the resulting multimodal representations are projected into the hidden space of the large language model to generate the final radiology report.

D. Knowledge Agent

The Knowledge Agent complements the graph branch with unstructured medical textual evidence. Specifically, we construct a retrieval corpus based on the PubMedVision dataset by extracting visual question-answering samples related to chest X-rays (CXR). Each question-answer pair is then summarized by a large language model into a structured knowledge item. These knowledge items are sequentially collected as knowledge snippets for subsequent retrieval.

Different from conventional similar-report retrieval methods, the external knowledge base used in this work is collected and organized from online medical resources, and is further summarized by a large language model into compact entity-level knowledge entries. This design offers two advantages. First, the retrieval query is explicitly grounded in the medical entities appearing in the current report, which reduces irrelevant evidence introduced by global case-level similarity or long-text retrieval. Second, compared with raw web text or complete reports, the retrieved knowledge entries are more focused and contain less redundancy, thereby providing clearer clinical evidence for key medical entities.

For each extracted entity e_i , the retrieval query is constructed from the complete preliminary report, the target entity, and its entity type:

$$\mathbf{q}_i = [R_0; e_i; t_i]. \quad (5)$$

The entity-wise retriever first encodes $\mathbf{q}_i$ using BGE-M3 and retrieves the top- N candidate chunks from a cached vector

database according to inner-product similarity. The candidate chunks are then re-ranked by a BGE reranker, and the top- K_r results are retained for each entity. In the current experiments, we set $N = 10$ and retain one re-ranked chunk for each entity.

Since different entities may retrieve the same supporting text, the retrieved chunks are merged and deduplicated according to their original chunk indices. Let:

$$\mathcal{C} = \{c_1, \dots, c_{N_r}\}, \quad (6)$$

denote the deduplicated evidence list. Each chunk is encoded by a frozen Bio_ClinicalBERT encoder with mean pooling:

$$\mathbf{r}_j = \text{Normalize} \left(\frac{\sum_{\ell} a_j^{\ell} \mathbf{H}_j^{\ell}}{\sum_{\ell} a_j^{\ell}} \right), \quad (7)$$

where $\mathbf{H}_j^{\ell}$ denotes the hidden state of the ℓ -th token in the j -th chunk c_j , and a_j^{ℓ} denotes the corresponding attention mask. The retrieved tokens are then stacked and padded within each batch:

$$\mathbf{R} = [\mathbf{r}_1; \dots; \mathbf{r}_{N_r}] \in \mathbb{R}^{N_r \times 768}. \quad (8)$$

The retrieval branch provides descriptive textual evidence that may not be explicitly represented in the triples of the graph branch. Encoding the retrieved chunks into continuous vectors avoids directly concatenating long textual prompts, thereby preserving entity-relevant external knowledge while keeping the input to the LLM compact.

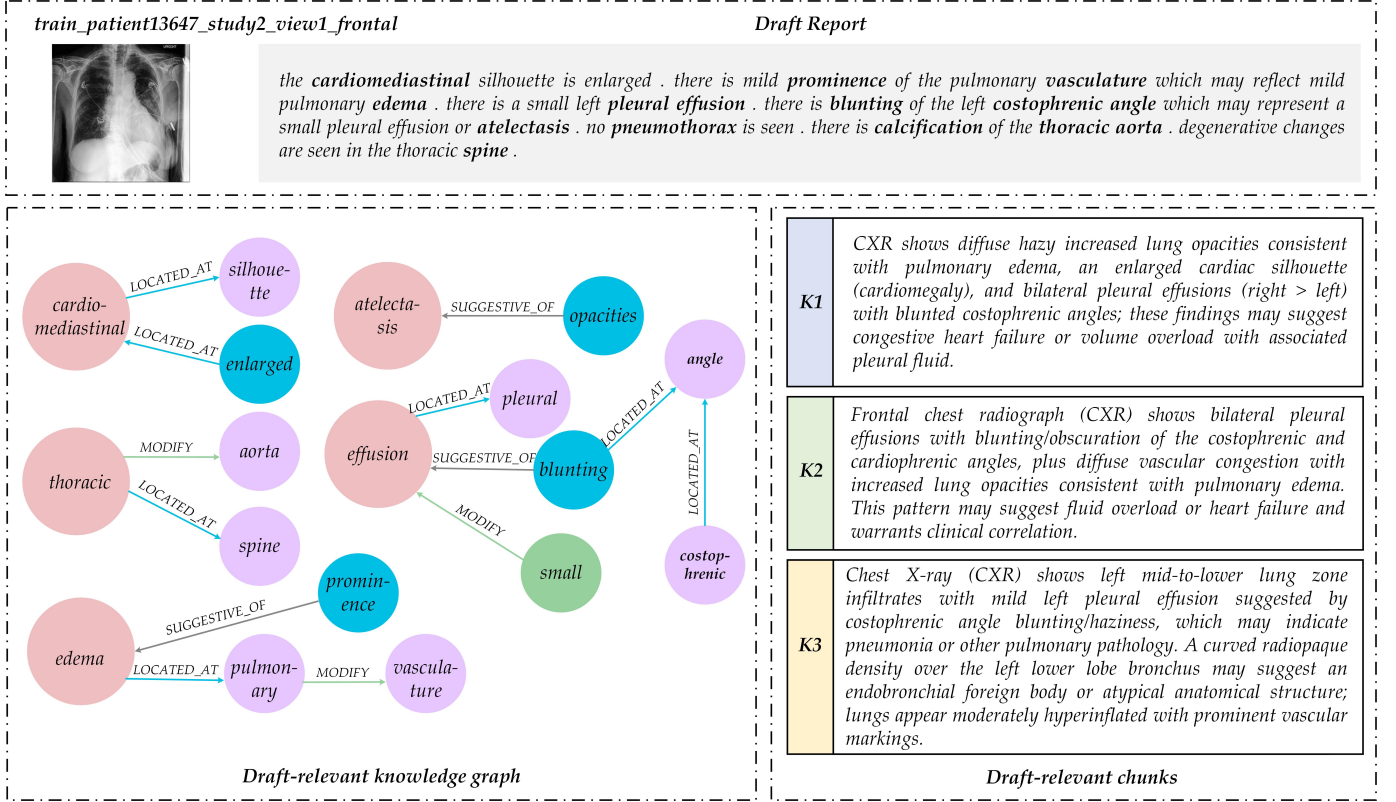


Fig. 3. The upper panel shows the input chest X-ray image and its corresponding Draft Report. The lower-left panel presents representative Draft-relevant knowledge graph triples retrieved by the MM-KG Agent and organized into five independent entity-centered subgraphs. The lower-right panel shows representative Draft-relevant text chunks retrieved by the Knowledge Agent. The two agents provide complementary structured and descriptive evidence for subsequent knowledge fusion and report refinement.

E. knowledge Fusion Module

The graph branch and the retrieval branch provide complementary knowledge. However, directly feeding such knowledge into the large language model would result in an excessively long prompt with substantial redundancy, which may limit its effectiveness in enhancing the model. Therefore, we introduce a knowledge fusion module.

We first fuse the knowledge from the graph branch. Specifically, the previously obtained embeddings h_i , h_{ij} , and q_{ij} are fed into a relation-aware graph attention encoder. For each attention head, the central entity embedding is transformed into a query, while the neighbor key is obtained by adding the corresponding edge embedding:

$$Q_i = W_Q h_i, \quad K_{ij} = W_K h_{ij} + q_{ij}, \quad U_{ij} = W_V h_{ij}.$$

The attention weight assigned to the j -th neighbor is computed as:

$$\alpha_{ij} = \frac{\exp(Q_i K_{ij}^\top / \sqrt{d})}{\sum_{\ell=1}^{K_i} \exp(Q_i K_{i\ell}^\top / \sqrt{d})}.$$

The graph contextual representation of entity e_i is then obtained by:

$$g_i = \text{MLP} \left(W_O \sum_{j=1}^{K_i} \alpha_{ij} U_{ij} \right).$$

The outputs of all entities are concatenated and dynamically padded to form the graph knowledge tokens:

$$G = [g_1; \dots; g_M] \in \mathbb{R}^{N_g \times 768}.$$

For the retrieval branch, as described above, the retrieved results are first merged and deduplicated. Each resulting chunk is then encoded using Bio-ClinicalBERT to obtain the retrieval-based knowledge representation R .

Finally, to fuse the knowledge from the graph and retrieval branches, we first project both representations into the same dimensional space as the visual features:

$$\tilde{G} = W_g G, \quad \tilde{R} = W_r R.$$

Let the image tokens output by the visual encoder be denoted as $V \in \mathbb{R}^{N_v \times d_v}$. The final multi-source knowledge representation is obtained through direct concatenation:

$$Z = [V; \tilde{R}; \tilde{G}].$$

The fused representation Z is projected into the hidden space of the large language model and normalized:

$$E = \text{LayerNorm}(W_{lm} Z).$$

Finally, E , together with the textual generation prompt, is used as the input prefix of the large language model for radiology report generation.

F. Loss Function

MAC-RRG is optimized using an autoregressive language modeling objective. After integrating the visual tokens, graph knowledge tokens, and retrieval knowledge tokens, the fused multimodal representation E is combined with the textual generation prompt embedding P to construct the conditioning prefix:

$$H = \text{Concat}(E, P). \quad (9)$$

Let $Y = (y_1, y_2, \dots, y_T)$ denote the ground-truth radiology report, where T is the number of report tokens. Conditioned on the multimodal prefix H , the probability of the target report is factorized in an autoregressive manner as

$$p_\theta(Y | H) = \prod_{t=1}^T p_\theta(y_t | y_{<t}, H), \quad (10)$$

where $y_{<t}$ denotes the report tokens preceding y_t , and θ represents the trainable model parameters.

The language modeling objective is defined as the average negative log-likelihood over the target report tokens:

$$\mathcal{L}_{\text{LM}} = -\frac{1}{T} \sum_{t=1}^T \log p_\theta(y_t | y_{<t}, H). \quad (11)$$

During training, the labels corresponding to the textual prompt, visual tokens, graph knowledge tokens, and retrieval knowledge tokens are masked with -100 , such that only the ground-truth radiology report tokens contribute to the language modeling loss.

IV. EXPERIMENTS

A. Datasets and Evaluation Metric

In our experiments, we employ three benchmark datasets that are widely used in the field of chest X-ray report generation, namely IU-Xray [52], MIMIC-CXR [53], and CheXpert Plus [54], to evaluate the effectiveness of the proposed MAC-RRG model. The linguistic quality of the generated reports is assessed using commonly adopted natural language generation (NLG) metrics, including BLEU [55], ROUGE-L [56], METEOR [57], and CIDEr [58]. In addition, following the evaluation protocol of R2Gen [9], we employ Clinical Efficacy (CE) metrics to evaluate the clinical accuracy of the generated reports. Detailed descriptions of the datasets and evaluation metrics are provided below.

- **IU-Xray Dataset.** IU-Xray is a publicly available medical imaging dataset consisting primarily of chest X-ray images and their corresponding radiology reports. Released by the Indiana University School of Medicine, the dataset contains 7,470 chest X-ray images and 3,955 associated radiology reports. To ensure comparability with existing methods, we follow the experimental settings adopted by R2Gen [9] and R2GenGPT [14], and partition the dataset into training, testing, and validation sets at a ratio of 7:1:2.

- **CheXpert Plus Dataset.** CheXpert Plus is a large-scale multimodal dataset comprising both medical images and textual radiology reports. It was developed to improve the performance, robustness, and fairness of machine learning

models in radiology. The dataset contains 223,228 chest X-ray images together with their corresponding reports and provides annotations for 14 categories of thoracic abnormalities, thereby offering high-quality data for research on medical image analysis and radiology report generation. CheXpert Plus has been applied to a variety of tasks, including disease diagnosis, medical image classification, abnormality labeling, and radiology report generation. To ensure fair and reproducible evaluation, we adopt the dataset partitioning protocol proposed in CXPMRG-Bench [16].

- **MIMIC-CXR Dataset.** MIMIC-CXR (Medical Information Mart for Intensive Care Chest X-ray) is a large-scale chest X-ray dataset jointly developed by the Massachusetts Institute of Technology and Beth Israel Deaconess Medical Center. It contains 377,110 chest X-ray images and 227,835 radiology reports. To enable a fair comparison with existing radiology report generation methods, we follow the experimental settings of R2Gen [9] and R2GenGPT [14], and use the official MIMIC-CXR splits for model training, validation, and testing.

For the evaluation of linguistic generation quality, BLEU measures the textual similarity between generated and reference reports based on n -gram precision. ROUGE-L evaluates content overlap by computing the longest common subsequence (LCS) between a generated report and its reference report. METEOR extends lexical matching by accounting for morphological variations, synonymous expressions, and differences in word order, thereby alleviating some of the limitations of BLEU in evaluating semantic consistency. CIDEr employs TF-IDF-weighted n -gram matching to emphasize informative textual expressions while reducing the excessive influence of frequently occurring phrases on the evaluation results.

In addition to linguistic evaluation, we employ Precision, Recall, and F1 score to assess the ability of MAC-RRG to recognize diseases, lesions, and other clinically relevant abnormalities. Precision measures the proportion of correctly identified positive clinical labels among all labels predicted as positive; therefore, a higher precision indicates fewer false-positive predictions. Recall measures the proportion of actual positive clinical labels that are successfully identified by the model, with a higher recall indicating that the model captures a larger proportion of clinically relevant abnormalities. The F1 score is the harmonic mean of Precision and Recall and thus provides a comprehensive measure of the model's overall performance in clinical abnormality recognition.

B. Implementation Details

We use a pre-trained Swin Transformer [60] as the visual encoder and Llama2-7B [61] as the language backbone. Contextual entities extracted from the initial draft report are used to query both the medical knowledge graph and the external chunk memory. For the MM-KG Agent, each entity retrieves up to 10 KG neighboring nodes, and the center nodes, neighboring nodes, and relation edges are encoded into 768-dimensional features using frozen Bio ClinicalBERT [62]. For the Knowledge Agent, each entity retrieves candidate chunks from a chunk memory containing 393 textual knowledge entries, followed by reranking. Then, the KG features

TABLE I
COMPARISON OF OUR MODEL’S PERFORMANCE ON IU X-RAY AND CHEXPRT PLUS DATASETS. THE BEST RESULT IS HIGHLIGHTED IN BOLD.

Dataset	Methods	Publication	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE-L	METEOR	CIDEr
IU X-Ray	R2Gen [9]	EMNLP 2020	0.470	0.304	0.219	0.165	0.371	0.187	-
	SentSAT+KG [31]	AAAI 2020	0.441	0.291	0.203	0.147	0.367	-	-
	R2GenCMN [10]	ACL-IJCNLP 2021	0.475	0.309	0.222	0.170	0.375	0.191	-
	PPKED [11]	CVPR 2021	0.483	0.315	0.224	0.168	0.376	0.187	0.351
	AlignTrans [32]	MICCAI 2021	0.484	0.313	0.225	0.173	0.379	0.204	-
	CMCL [33]	ACL 2021	0.473	0.305	0.217	0.162	0.378	0.186	-
	DCL [34]	CVPR 2023	-	-	-	0.163	0.383	0.193	0.586
	R2GenGPT [14]	Meta Radiology 2023	0.465	0.299	0.214	0.161	0.376	0.219	0.542
	PromptMRG [35]	AAAI 2024	0.401	-	-	0.098	0.160	0.281	-
	SILC [36]	IEEE TMI 2024	0.472	0.321	0.234	0.175	0.379	0.192	0.368
	MMR [37]	BSPC 2025	0.497	0.333	0.240	0.185	0.399	0.215	-
	DuCo-Net [38]	IEEE Access 2025	0.500	0.330	0.220	0.160	0.260	0.240	-
	DVAF-DMSR [39]	AAAI 2026	0.495	0.328	0.242	0.187	0.384	-	0.634
	DVPAAlign [40]	ICASSP 2026	0.431	-	-	0.110	0.318	0.166	-
	MAC-RRG	Ours	0.488	0.330	0.249	0.197	0.400	0.223	0.703
CheXpert Plus	R2Gen [9]	EMNLP 2020	0.301	0.179	0.118	0.081	0.246	0.113	0.077
	R2GenCMN [10]	ACL-IJCNLP 2021	0.321	0.195	0.128	0.087	0.256	0.127	0.102
	XProNet [41]	ECCV 2022	0.364	0.225	0.148	0.100	0.265	0.146	0.121
	ORGan [42]	ACL 2023	0.320	0.196	0.128	0.086	0.261	0.135	0.107
	R2GenGPT [14]	Meta Radiology 2023	0.361	0.224	0.149	0.101	0.266	0.145	0.123
	ASGMD [43]	ESWA 2024	0.267	0.149	0.094	0.063	0.220	0.094	0.044
	Token-Mixer [44]	IEEE TMI 2024	0.378	0.231	0.153	0.091	0.262	0.135	0.098
	PromptMRG [35]	AAAI 2024	0.326	0.174	-	0.095	0.222	0.121	0.044
	MCA-RG [45]	MICCAI 2025	0.367	0.218	0.149	0.102	0.266	0.147	-
	VLCI [46]	IEEE TIP 2025	-	-	-	0.080	0.247	-	0.072
	DVPAAlign [40]	ICASSP 2026	0.335	-	-	0.093	0.228	0.127	-
	MAC-RRG	Ours	0.375	0.232	0.154	0.105	0.267	0.151	0.121

TABLE II
COMPARISON OF OUR MODEL’S PERFORMANCE ON THE MIMIC-CXR DATASETS. THE BEST RESULT IS HIGHLIGHTED IN BOLD.

Dataset	Methods	Publication	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE-L	METEOR	CIDEr
MIMIC-CXR	R2Gen [9]	EMNLP 2020	0.353	0.218	0.145	0.103	0.277	0.142	-
	R2GenCMN [10]	ACL-IJCNLP 2021	0.353	0.218	0.148	0.106	0.278	0.142	-
	PPKED [11]	CVPR 2021	0.360	0.224	0.149	0.106	0.284	0.149	0.237
	AlignTrans [32]	MICCAI 2021	0.378	0.235	0.156	0.112	0.283	0.158	-
	CMCL [33]	ACL 2021	0.344	0.217	0.140	0.097	0.281	0.133	-
	Clinical-BERT [47]	AAAI 2022	0.383	0.230	0.151	0.106	0.275	0.144	0.151
	METransformer [12]	CVPR 2023	0.386	0.250	0.169	0.124	0.291	0.152	0.362
	DCL [34]	CVPR 2023	-	-	-	0.109	0.284	0.150	0.281
	R2GenGPT [14]	Meta Radiology 2023	0.405	0.252	0.171	0.123	0.285	0.167	0.254
	PromptMRG [35]	AAAI 2024	0.398	-	-	0.112	0.268	0.157	-
	AdaMatch-Cyclic [48]	ACL 2024	0.379	0.235	0.154	0.106	0.286	0.163	-
	DACG [49]	MIA 2025	0.398	0.249	0.167	0.117	0.290	0.162	-
	Teaser [50]	IEEE TMI 2025	0.423	0.257	0.166	0.113	0.287	0.170	-
	GDMRG [51]	arXiv 2026	0.406	-	-	0.114	0.272	0.157	-
	DVPAAlign [40]	ICASSP 2026	0.408	-	-	0.118	0.276	0.164	-
	MAC-RRG	Ours	0.409	0.258	0.177	0.128	0.285	0.165	0.248

TABLE III
A COMPARISON OF THE CLINICAL EFFICACY (CE) METRICS BETWEEN OUR PROPOSED FRAMEWORK (OURS) AND STATE-OF-THE-ART METHODS USING F1 SCORE, PRECISION, AND RECALL ON THE CHEXPRT PLUS DATASET.

Model	F1	Precision	Recall
R2Gen [9]	0.181	0.318	0.200
R2GenCMN [10]	0.231	0.329	0.241
WCL [59]	0.256	0.335	0.259
PromptMRG [35]	0.281	0.258	0.265
R2GenGPT [14]	0.260	0.315	0.224
ORGan [42]	0.277	0.288	0.287
Token-Mixer [44]	0.288	0.309	0.270
VLCI [46]	0.163	0.341	0.175
MAC-RRG (Ours)	0.295	0.350	0.274

and retrieval features are fused and mapped to the 1024-dimensional visual feature space. They are concatenated with visual tokens and further projected into the 4096-dimensional LLM embedding space for report generation. The model is implemented in PyTorch [63] and optimized with AdamW [64] using a learning rate of 1×10^{-4} . A cosine annealing scheduler is adopted with a minimum learning rate of 1×10^{-6} . All experiments are conducted on an NVIDIA A800SXM4-80GB GPU. More details can be found in our source code.

C. Comparison on Public Benchmark Datasets

• **Analysis of the NLG Metrics.** To comprehensively evaluate the effectiveness of the proposed method for medical image report generation, we conduct comparative experiments on

TABLE IV

ABLATION STUDY ON CHEXPRT PLUS DATASET, ASSESSING THE IMPACT OF KEY COMPONENTS: MM-KG AGENT (MM-KG), KNOWLEDGE AGENT (KA), KNOWLEDGE FUSION MODULE (KFM). A “✓” INDICATES THE PRESENCE OF EACH COMPONENT, WHILE “-” DENOTES ITS ABSENCE.

Dataset	Setting	MM-KG	KA	KFM	BLEU-1	BLEU-2	BLEU-3	BLEU-4	RG-L	METEOR	CIDEr	F1	Precision	Recall
CheXpert Plus	BASE	-	-	-	0.361	0.224	0.149	0.101	0.266	0.145	0.123	0.260	0.315	0.224
	(a)	✓	-	-	0.366	0.227	0.150	0.103	0.266	0.148	0.124	0.286	0.340	0.270
	(b)	-	✓	-	0.370	0.228	0.151	0.103	0.266	0.149	0.125	0.288	0.344	0.269
	(c)	✓	✓	-	0.372	0.231	0.153	0.104	0.265	0.150	0.122	0.292	0.359	0.270
	(d)	✓	✓	✓	0.375	0.232	0.154	0.105	0.267	0.151	0.121	0.295	0.350	0.274

three widely used public benchmark datasets, including IU X-Ray, CheXpert Plus, and MIMIC-CXR. We adopt commonly used natural language generation evaluation metrics, including BLEU, ROUGE-L, METEOR, and CIDEr. The compared methods cover traditional encoder-decoder frameworks, cross-modal alignment methods, pre-trained language model enhanced methods, and knowledge-/graph-enhanced methods, thereby providing a comprehensive evaluation of the report generation capability of the proposed method on datasets with different scales and complexities.

As shown in Table I, on the IU X-Ray dataset, the proposed MAC-RRG achieves competitive overall performance, with BLEU-1, BLEU-2, BLEU-3, BLEU-4, ROUGE-L, METEOR, and CIDEr scores of 0.488, 0.330, 0.249, 0.197, 0.400, 0.223, and 0.703, respectively. Specifically, MAC-RRG achieves the best results on BLEU-3, BLEU-4, ROUGE-L, and CIDEr. This indicates that the proposed method not only maintains good word-level matching capability, but also has stronger ability in generating continuous medical phrases, complex semantic fragments, and complete report contents.

On the more complex CheXpert Plus dataset, MAC-RRG also demonstrates stable and excellent generation performance. As shown in Table I, the proposed method achieves BLEU-1, BLEU-2, BLEU-3, BLEU-4, ROUGE-L, METEOR, and CIDEr scores of 0.375, 0.232, 0.154, 0.105, 0.267, 0.151, and 0.121, respectively. Among these metrics, MAC-RRG achieves the best performance on BLEU-2, BLEU-3, BLEU-4, ROUGE-L, and METEOR, indicating that the proposed method can better capture semantic dependencies among medical entities and generate medical descriptions that are highly consistent with the reference reports. Although the BLEU-1 score is slightly lower than that of Token-Mixer, which achieves 0.378, and the CIDEr score is slightly lower than that of R2GenGPT, which achieves 0.123, the gaps are relatively small. This suggests that the proposed method maintains strong text matching capability while paying more attention to the consistency of medical semantic relations, contextual structure, and overall report quality.

On the larger-scale MIMIC-CXR dataset with more complex clinical descriptions, MAC-RRG further demonstrates good generalization ability and stability. As shown in Table II, the proposed method achieves BLEU-1, BLEU-2, BLEU-3, BLEU-4, ROUGE-L, METEOR, and CIDEr scores of 0.409, 0.258, 0.177, 0.128, 0.285, 0.165, and 0.248, respectively. Specifically, MAC-RRG obtains the best results on BLEU-2, BLEU-3, and BLEU-4, showing its strong advantages in generating continuous medical phrases, disease-related descriptive fragments, and complex sentence structures. Although it

does not achieve the highest scores on some metrics such as BLEU-1, ROUGE-L, METEOR, and CIDEr, its overall performance remains highly competitive. In particular, the advantages on high-order BLEU metrics demonstrate that the proposed method can generate report contents that are more consistent with medical writing conventions and semantically more coherent.

Overall, MAC-RRG achieves competitive NLG performance on the IU X-Ray, CheXpert Plus, and MIMIC-CXR datasets. On the relatively small-scale IU X-Ray dataset, the proposed method shows prominent performance in high-order semantic matching and report content consistency. On the more complex CheXpert Plus and MIMIC-CXR datasets, the proposed method still maintains stable generation quality and achieves the best or near-best results on multiple key metrics. These experimental results fully verify the effectiveness and robustness of the proposed method across datasets with different scales and clinical complexities. They also demonstrate that the knowledge-enhanced strategy based on graph path retrieval and knowledge base retrieval can help the model generate more accurate, coherent, and medically reasonable automated radiology reports.

• **Analysis of CE Metric.** As shown in Table III, we further evaluate the clinical correctness of different medical report generation models on the CheXpert Plus dataset using Clinical Efficacy (CE) metrics, including F1 score, Precision, and Recall. These metrics measure whether the generated reports can accurately capture clinically relevant abnormalities, which is crucial for medical report generation. Compared with existing methods, the proposed MAC-RRG achieves the best results in terms of F1 score and Precision, reaching 0.295 and 0.350, respectively. Specifically, MAC-RRG improves the F1 score from 0.260 achieved by R2GenGPT to 0.295. Meanwhile, its Precision also surpasses that of VLCI, which obtains the highest Precision among the baseline methods. This indicates that our method is more effective in generating clinically accurate pathological findings and reducing false-positive descriptions in the generated reports. Although OR-Gan achieves the highest Recall of 0.287, its Precision and F1 score are only 0.288 and 0.277, respectively, which are clearly lower than those of MAC-RRG. In contrast, our method achieves a competitive Recall of 0.274 while maintaining the highest Precision and overall F1 score. This demonstrates that MAC-RRG achieves a better balance between correctly identifying pathological findings and avoiding inaccurate clinical statements. Overall, the CE metric results demonstrate that the proposed method can effectively identify key clinical abnormalities from medical images and generate reports with

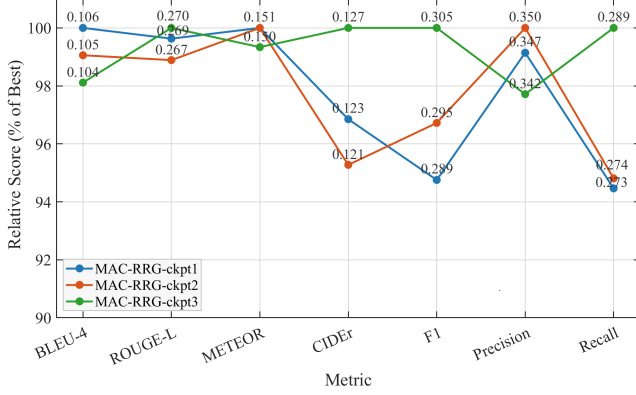


Fig. 4. Comparison of different checkpoints on BLEU-4, ROUGE-L, METEOR, CIDEr, F1, Precision, and Recall.

stronger clinical consistency. The higher F1 score further verifies the effectiveness of MAC-RRG in balancing sensitivity and precision, indicating its potential for clinically reliable automatic medical report generation.

D. Ablation Study

• **Component Analysis.** To further validate the effectiveness of each key component in the proposed MAC-RRG, we conduct component analysis experiments on the CheXpert Plus dataset. As shown in Table IV, we mainly evaluate the contributions of three core components, including the MM-KG Agent, the Knowledge Agent, and the Knowledge Fusion Module. Specifically, the MM-KG Agent introduces structured radiological relations by expanding extracted medical entities into local knowledge graph neighborhoods, while the Knowledge Agent retrieves entity-level textual evidence from an external medical knowledge base. The Knowledge Fusion Module further integrates these two complementary knowledge sources into compact representations, reducing the redundancy of long textual prompts and improving knowledge utilization during report generation.

In terms of NLG metrics, both the MM-KG Agent and the Knowledge Agent consistently improve over the baseline. Specifically, the baseline achieves BLEU-1, BLEU-2, BLEU-3, BLEU-4, ROUGE-L, METEOR, and CIDEr scores of 0.361, 0.224, 0.149, 0.101, 0.266, 0.145, and 0.123, respectively. With only the MM-KG Agent, these scores improve to 0.366, 0.227, 0.150, 0.103, 0.266, 0.148, and 0.124, while using only the Knowledge Agent yields 0.370, 0.228, 0.151, 0.103, 0.266, 0.149, and 0.125. When all three components are jointly used, the model achieves the best overall performance on most NLG metrics, with BLEU-1, BLEU-2, BLEU-3, BLEU-4, ROUGE-L, and METEOR scores further improved to 0.375, 0.232, 0.154, 0.105, 0.267, and 0.151, respectively. This suggests that the collaboration of structured knowledge encoding, entity-level knowledge retrieval, and the knowledge fusion module can effectively improve the linguistic quality and semantic consistency of generated reports. Similar improvements are also observed in clinical efficacy metrics,

TABLE V
COMPARE THE EFFECTS OF DIFFERENT NUMBERS OF NEIGHBORS.

#top_K	BLEU-4	ROUGE-L	METEOR	CIDEr
3	0.102	0.269	0.147	0.131
5	0.103	0.267	0.148	0.128
10	0.105	0.267	0.151	0.121
15	0.102	0.266	0.146	0.129
20	0.103	0.269	0.148	0.131

where F1, Precision, and Recall increase from 0.260, 0.315, and 0.224 in the baseline to 0.295, 0.350, and 0.274 in the full model. These results demonstrate that the MM-KG Agent and the Knowledge Agent provide complementary structured and textual medical knowledge, while the Knowledge Fusion Module effectively integrates them, thereby improving the linguistic quality, semantic consistency, and clinical reliability of generated reports.

• **Analysis of Checkpoint Selection Strategy.** As shown in Fig. 4, the final MAC-RRG model is selected based on a balanced consideration of BLEU-4 and clinically relevant Precision, Recall, and F1, rather than the best CIDEr score alone. Specifically, MAC-RRG-ckpt1 achieves the highest BLEU-4 score of 0.106, but its F1 and Recall are relatively lower at 0.289 and 0.273. MAC-RRG-ckpt3 obtains the best CIDEr, ROUGE-L, F1, and Recall scores of 0.127, 0.270, 0.305, and 0.289, respectively, but its BLEU-4 and Precision decrease to 0.104 and 0.342, indicating a tendency toward broader yet less precise clinical descriptions. By comparison, MAC-RRG-ckpt2 achieves a competitive BLEU-4 score of 0.105 and the highest Precision of 0.350, while also outperforming ckpt1 in F1 and Recall. Therefore, we adopt MAC-RRG-ckpt2 as the final model, since it provides a better trade-off between language quality and clinical factual accuracy.

• **Analysis on different numbers of neighbors.** To analyze the influence of different neighborhood scales in the knowledge graph, we conduct ablation experiments by varying the number of retrieved neighboring nodes. As shown in Table V, when top_K ranges from 3 to 20, the overall NLG performance remains relatively stable, with BLEU-4, ROUGE-L, METEOR, and CIDEr varying within 0.102–0.105, 0.266–0.269, 0.146–0.151, and 0.121–0.131, respectively. Increasing top_K from 3 to 10 improves BLEU-4 from 0.102 to 0.105 and METEOR from 0.147 to 0.151, indicating that a moderate number of neighbors can provide useful structured relational information. When top_K is further increased to 15, BLEU-4 and METEOR decrease to 0.102 and 0.146, respectively. At $\text{top}_K=20$, BLEU-4 and METEOR slightly recover to 0.103 and 0.148, while ROUGE-L and CIDEr reach relatively high values of 0.269 and 0.131. These results suggest that although more neighbors may increase knowledge coverage, they may also introduce redundant or weakly relevant information. Overall, $\text{top}_K=10$ achieves the best BLEU-4 and METEOR scores while maintaining competitive ROUGE-L performance, and is therefore adopted in the final model.

• **Analysis of contextual entity types.** As shown in Table VI, different entity types have different effects on report generation performance. When only anatomy entities are used, the model

TABLE VI
COMPARE THE EFFECTS OF CONTEXTUAL ENTITY TYPES.

#entity_types	BLEU-4	ROUGE-L	METEOR	CIDEr
A	0.102	0.269	0.148	0.133
A + D	0.105	0.267	0.151	0.121
A + D+ Dev	0.103	0.266	0.149	0.115
All	0.103	0.265	0.147	0.120

TABLE VII
COMPARE THE EFFECTS OF DIFFERENT NUMBERS OF CHUNKS.

#chunks	BLEU-4	ROUGE-L	METEOR	CIDEr
200	0.104	0.266	0.150	0.121
300	0.104	0.266	0.148	0.122
400	0.105	0.267	0.151	0.121
1000	0.106	0.266	0.150	0.123
1500	0.103	0.263	0.146	0.121

achieves the highest CIDEr score of 0.133, indicating that anatomy entities provide stable spatial localization information. After introducing disorder entities, the A + D setting obtains the best BLEU-4 and METEOR scores, reaching 0.105 and 0.151, respectively, improving over the anatomy-only setting with 0.102 and 0.148. This suggests that disease-related entities help enhance the description of abnormal findings. In contrast, further adding device entities reduces CIDEr from 0.121 to 0.115. When all entity types are used, ROUGE-L and METEOR also decrease to 0.265 and 0.147, respectively, indicating that excessive entity types may introduce noise or weakly relevant information. Therefore, we adopt Anatomy + Disorder as the default entity-type setting to achieve a better balance between anatomical localization and lesion-level semantic description.

• **Analysis of different numbers of chunks.** We investigate the effect of retrieval corpus size by varying the nominal number of knowledge chunks from 200 to 1,500 while keeping all other settings unchanged. After preprocessing and deduplication, the nominal 400-chunk configuration contains 393 valid chunks. As shown in Table VII, increasing the corpus size from 200 to 400 generally improves generation performance, with the 400-chunk setting achieving the best or comparable ROUGE-L and METEOR scores and competitive BLEU-4 and CIDEr results. Checkpoint selection jointly considers NLG and clinical efficacy metrics rather than BLEU-4 alone. Although another checkpoint under the 400-chunk setting also reaches a BLEU-4 score of 0.106, it is not selected because of its lower Precision, Recall, and F1 scores. Similarly, the 1,000-chunk setting achieves a BLEU-4 score of 0.106 but obtains only 0.278, 0.334, and 0.263 in F1, Precision, and Recall, respectively, lower than the selected 400-chunk checkpoint (0.295, 0.350, and 0.274). Moreover, increasing the corpus size to 1,500 chunks degrades all NLG metrics, possibly due to redundant or weakly relevant evidence. Therefore, we adopt the nominal 400-chunk configuration to balance linguistic quality and clinical efficacy.

• **Analysis of different fusion strategies.** The final version of this study adopts direct concatenation as the knowledge fusion

TABLE VIII
COMPARE THE EFFECTS OF DIFFERENT FUSION STRATEGIES

#strategies	BLEU-4	ROUGE-L	METEOR	CIDEr
self_attention	0.102	0.266	0.148	0.125
concat	0.105	0.267	0.151	0.121
cross_attention	0.104	0.267	0.149	0.116

strategy, rather than introducing additional fusion through self-attention or cross-attention, mainly based on the experimental results. As shown in Table VIII, the concat strategy achieves the best performance on BLEU-4 and METEOR, while maintaining competitive results on ROUGE-L. In contrast, self-attention and cross-attention do not bring stable improvements.

This indicates that the graph branch and the retrieval branch have already formed effective continuous knowledge representations after graph attention encoding and Bio-ClinicalBERT encoding. Introducing additional complex attention interactions may increase the number of parameters and amplify the influence of retrieval noise or weakly relevant neighboring nodes in the knowledge graph. Therefore, direct concatenation can preserve graph-structured knowledge, retrieved textual knowledge, and visual information more stably while maintaining a simple architecture, thereby enabling effective enhancement of the large language model.

E. Visualization

• **Visualization of Structured and Textual Knowledge Evidence.** As shown in Fig. 3, we select the chest X-ray image `train_patient13647_study2_view1_frontal` and its corresponding Draft Report as a representative case to illustrate how the proposed collaborative agents retrieve and organize complementary knowledge. The upper part of the figure presents the input chest radiograph together with the initial Draft Report, which describes cardiomeastinal enlargement, mild pulmonary vascular prominence with possible pulmonary edema, a small left pleural effusion, left costophrenic angle blunting with possible atelectasis, absence of pneumothorax, thoracic aortic calcification, and degenerative changes of the thoracic spine.

Given the Draft Report, the MM-KG Agent first identifies case-relevant contextual entities and retrieves their associated triples from the medical knowledge graph. For clarity and interpretability, only a subset of representative triples is visualized in the lower-left panel. These triples are organized into five independent entity-centered subgraphs corresponding to cardiomeastinal findings, atelectasis, pleural effusion, thoracic anatomical structures, and pulmonary edema. The relations `LOCATED_AT`, `MODIFY`, and `SUGGESTIVE_OF` encode anatomical associations, modifier relationships, and diagnostic associations between radiographic findings and potential abnormalities, respectively. Consequently, medical concepts that are dispersed throughout the Draft Report are re-organized into structured and traceable graph-based evidence.

In parallel, the Knowledge Agent retrieves semantically relevant text chunks from an external knowledge base, as shown in the lower-right panel. The representative chunks K1–

Image	Ground Truth	Ours	R2GenGPT
	Loculated pleural effusions are again seen. A rounded pleural-based opacity posteriorly laterally at the right lung apex also likely reflects a loculated effusion. Patchy opacities are also seen throughout the right lung zone, most prominent the right lung base, which may reflect adjacent atelectasis or consolidation. The left lung appears largely clear. The cardiomeastinal silhouette appears unchanged. No acute bony abnormalities are identified. Multilevel degenerative changes are again seen of the thoracic spine.	postsurgical changes of the right hemithorax are again seen. there is a small right pleural effusion with associated atelectasis or consolidation at the right base. the left lung is clear. the heart is normal in size. the mediastinal contours are normal. no acute osseous abnormalities are identified. degenerative changes of the thoracic spine are again seen.	the cardiomeastinal silhouette is unchanged. there is redemonstration of a small right apical pneumothorax which has decreased in size since the prior examination. there is persistent elevation of the right hemidiaphragm with associated blunting of the right costophrenic angle. there is no evidence of pneumothorax or pleural effusion on the left side. there are no osseous abnormalities. surgical clips are again seen in the right upper quadrant.
	The cardiac silhouette is enlarged. Atherosclerotic calcifications of the aorta. Prominent vascular markings consistent with pulmonary edema. Blunting of the bilateral costophrenic angles, likely small pleural effusions. Crowding of the vascular markings in the bilateral bases. Osteopenia.	single ap view of the chest demonstrates diffuse increased interstitial markings consistent with pulmonary edema. there are small bilateral pleural effusions. in addition there is blunting of the costophrenic angles consistent with small bilateral pleural effusions. the cardiomeastinal silhouette is unremarkable. there is atherosclerotic calcification of the aortic arch and thoracic aorta. bones demonstrate osteopenia. no soft tissue abnormalities are identified.	the cardiomeastinal silhouette is within normal limits. there are bibasilar opacities which may represent atelectasis or consolidation. there are also bilateral pleural effusions. the visualized osseous structures are unremarkable. no acute bony abnormalities are identified. degenerative changes are seen in the thoracic spine.
	Cardiac silhouette is not enlarged. Mediastinal contour is normal. Multiple calcified mediastinal and hilar nodes. Abnormal hazy and nodular opacities at both lung bases are concerning for pneumonia or aspiration. Possible emphysema. No acute bony abnormalities. No pneumothorax.	the cardiomeastinal silhouette is normal. increased interstitial opacities are noted at the bilateral lung bases. no pleural effusions. no pneumothorax. multilevel degenerative changes of the thoracic spine. surgical clips are seen in the right axilla. there are no acute osseous abnormalities. biapical pleural thickening is noted on the lateral radiograph.	pa and lateral chest radiographs are obtained. the cardiomeastinal silhouette is within normal limits. there is mild prominence of the pulmonary vasculature. there is no focal consolidation or pleural effusion. there is blunting of the costophrenic angles bilaterally which may represent small pleural effusions. degenerative changes are seen in the thoracic spine.

Fig. 5. X-ray images and their corresponding ground-truth reports, along with the generated reports of MAC-RRG and R2GenGPT on the CheXpert Plus dataset. Matching sentences in our report are highlighted in yellow, R2GenGPT matching sentences are highlighted in cyan, and sentences matched by both models are highlighted in pink.

K3 contain radiological descriptions related to cardiomegaly, pulmonary vascular congestion or pulmonary edema, pleural effusion, costophrenic angle blunting, and lower-lung abnormalities. In contrast to the concise relational evidence provided by the MM-KG Agent, these chunks supply richer descriptive context and broader co-occurrence patterns for the findings mentioned in the Draft Report.

Overall, the visualization demonstrates the complementary roles of the two knowledge agents. The MM-KG Agent provides structured evidence at the entity and relation levels, whereas the Knowledge Agent contributes descriptive textual evidence. These complementary knowledge sources jointly support the subsequent knowledge-fusion module and facilitate the refinement of the initial radiology report.

• **Report Generation.** As shown in Figure 5, we present qualitative examples from the CheXpert Plus dataset to further demonstrate the effectiveness of the proposed MAC-RRG model for X-ray image-based radiology report generation. For each X-ray image, we compare the ground-truth report with the reports generated by MAC-RRG and the baseline model R2GenGPT. To make the comparison more intuitive, sentences in the reports generated by our model that match the ground truth are highlighted in yellow, those generated by R2GenGPT that match the ground truth are highlighted in cyan, and sentences correctly matched by both models are highlighted in pink.

The visualization results show that MAC-RRG achieves stronger consistency with the ground-truth reports in both positive findings and negative observations. This indicates that,

compared with R2GenGPT, the reports generated by MAC-RRG are more semantically aligned with the ground truth. These qualitative results further demonstrate that the proposed multi-agent collaborative framework can improve the clinical completeness and factual consistency of generated radiology reports.

F. Limitation Analysis

Although MAC-RRG improves radiology report generation by incorporating structured knowledge graph evidence and entity-level retrieved textual knowledge, its performance is still influenced by the quality of the initial draft report. Since contextual entities are extracted from the preliminary report, missed or incorrectly described abnormalities may lead the MM-KG Agent and Knowledge Agent to retrieve incomplete or less relevant external evidence. This dependency indicates that more robust entity extraction and draft correction mechanisms remain important for further improving iterative report refinement.

The current framework adopts direct concatenation as the final knowledge fusion strategy after projecting graph and retrieval representations into the visual feature space. We also investigated more complex fusion strategies, including self-attention and cross-attention. However, as reported in Table VIII, these strategies do not consistently improve the overall performance under the current setting. Self-attention achieves a slightly higher CIDEr score but reduces BLEU-4 and METEOR, while cross-attention obtains comparable ROUGE-L but leads to a lower CIDEr score. One possible

reason is that additional attention interactions may amplify weakly relevant retrieved chunks or noisy neighboring nodes in the knowledge graph, especially when external knowledge has already been encoded into compact continuous representations. Therefore, direct concatenation is adopted in this work as a more stable and lightweight fusion strategy.

V. CONCLUSION

In this paper, we propose MAC-RRG, an iterative multi-agent collaboration framework for X-ray radiology report generation. The core idea is to use the initial draft report as an entity-aware bridge between visual evidence and external medical knowledge. Specifically, the MM-KG agent retrieves structured entity-centered relations from a medical knowledge graph, while the Knowledge agent retrieves complementary entity-level textual evidence from an external knowledge corpus. These two knowledge sources are fused with visual tokens and projected into the LLM embedding space to guide the generation of more clinically consistent reports. Extensive experiments on IU X-Ray, CheXpert Plus, and MIMIC-CXR demonstrate the effectiveness of the proposed framework. MAC-RRG achieves competitive or superior performance on multiple NLG metrics, especially on high-order BLEU scores, indicating its ability to generate more coherent medical phrases and disease-related descriptions.

Overall, MAC-RRG shows that role-specialized knowledge agents can provide complementary structured and textual evidence for LLM-based radiology report generation. In future work, we will further investigate more robust entity extraction, larger-scale medical knowledge bases, uncertainty-aware retrieval, and adaptive knowledge fusion strategies to improve the generalization and clinical trustworthiness of automatic radiology report generation systems.

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